SVIT HackVerse 2K26

Duration 24th April 2026 to 25th April 2026

List of Problem Statements

Sr. No.	Unique ID	Problem Statement
1	**SVIT PS 01**	Build a multi-agent AI platform that autonomously ingests open-source intelligence (OSINT) from multiple data streams, detects early signs of conflict escalation, simulates strategic response scenarios with projected outcomes, models civilian impact, and delivers a single explainable commander-grade intelligence brief within 5 minutes of any new threat signal with every recommendation traceable to its source.
2	**SVIT PS 02**	Build a multi-agent AI system that passively collects information from company tools (emails, chats, documents, meetings), understands and stores the reasoning behind decisions not just what was decided and lets anyone ask plain English questions like "Why did we choose this vendor?" or "What breaks if we change this feature?" with answers that show exactly where the information came from.
3	**SVIT PS 03**	Build a multi-agent AI system that automatically monitors regulatory sources (RBI, SEBI, MCA), detects exactly what changed in new circulars compared to previous versions, maps those changes to a company's specific products, contracts, and internal policies, drafts the required amendments, and delivers a ready-to-use impact report within 24 hours of any new regulation being published.
4	**SVIT PS 04**	Build a multi-agent AI system that connects to a GitHub / GitLab, analyzes code change history, bug patterns, test coverage trends, and deployment frequency, assigns a health score to each part of the codebase, predicts which components are most likely to cause a failure or slowdown in the next 90 days, and generates a business-impact report written for a non-technical CEO showing the cost of inaction in time and money.
5	**SVIT PS 05**	Build a multi-agent SaaS platform that sits as an invisible observability layer on top of any deployed AI agent system autonomously monitors every agent decision, action, and outcome in real time, detects when agents are hallucinating, going off-task, or causing business harm, calculates verified ROI per agent per workflow, generates a compliance-ready audit trail for every autonomous action, and delivers a weekly governance brief to the CIO making AI agents trustworthy enough for regulated industries to actually deploy at scale.
6	**SVIT PS 06**	Contemporary deep learning systems rely on static neural architectures that are manually designed by domain experts, making them computationally expensive, rigid, and unable to adapt to dynamic or evolving data distributions. Neural Architecture Search (NAS) methods, while promising, remain largely offline, computationally prohibitive, and disconnected from real-time feedback loops. The absence of self-adaptation mechanisms forces practitioners to repeatedly redesign and retrain models from scratch when problem domains shift, resulting in

		significant resource wastage and performance degradation over time. Furthermore, existing meta-learning frameworks lack the ability to autonomously modify their own internal structure in response to task complexity, concept drift, or resource constraints. This thesis investigates the fundamental problem of enabling neural networks to evolve their own architecture autonomously through meta-learning strategies, without requiring human intervention or exhaustive search processes.
7	**SVIT PS 07**	Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language processing tasks; however, their internal decision-making processes remain largely opaque, rendering them unsuitable for high-stakes applications such as healthcare, legal reasoning, and financial advisory. The black-box nature of transformer-based generative models creates critical challenges related to accountability, bias detection, and regulatory compliance. Existing post-hoc explainability techniques, such as attention visualization and SHAP-based attribution, fail to provide causal and semantically meaningful explanations for generative outputs. Additionally, current interpretability methods are not natively integrated into the generation pipeline, leading to misalignment between model explanations and actual generative behavior. This research addresses the gap between model performance and trustworthiness by developing intrinsic explainability mechanisms that are embedded within the architecture of LLMs without significantly compromising their generative capacity.
8	**SVIT PS 08**	Smart systems including smart grids, autonomous vehicles, and intelligent healthcare platforms generate massive volumes of sensitive data that must be leveraged for real-time sequential decision-making. Traditional centralized deep reinforcement learning (DRL) approaches require aggregating private data at a central server, creating severe privacy risks, communication bottlenecks, and single-point-of-failure vulnerabilities. While federated learning offers a distributed data paradigm, its integration with reinforcement learning remains technically challenging due to non-stationarity of environments, heterogeneous reward structures across agents, and the convergence instability introduced by local policy gradient updates. Differential privacy mechanisms applied naively to federated RL systems introduce excessive noise that degrades policy performance. This thesis addresses the absence of a unified, privacy-preserving, and communication-efficient federated DRL framework capable of operating across heterogeneous smart system environments.
9	**SVIT PS 09**	Optimization problems arising in complex systems such as supply chain logistics, financial portfolio management, and protein folding exhibit combinatorial complexity that renders classical algorithms computationally intractable at scale. Quantum computing offers theoretical advantages through superposition and entanglement; however, current Noisy Intermediate-Scale Quantum (NISQ) devices suffer from decoherence, gate error rates, and limited qubit counts that prevent the realization of fully quantum optimization algorithms. Purely classical heuristics, on the other hand, fail to leverage quantum speedups and struggle with NP-hard problem instances. A critical research gap exists in designing hybrid quantum-classical optimization architectures that intelligently partition problem instances between quantum and classical components, adapt to device noise profiles, and deliver measurable speedups in real-time operational contexts. This thesis investigates algorithmic strategies for building robust, scalable, and practically deployable hybrid optimization systems.
10	**SVIT PS 10**	Contemporary blockchain-based identity management systems rely

		predominantly on elliptic curve cryptography (ECC) and RSA-based digital signatures, which are provably vulnerable to quantum attacks via Shor's algorithm. As quantum computers approach cryptographically relevant qubit thresholds, the long-term security of decentralized identity systems is fundamentally threatened. The transition to post-quantum cryptographic (PQC) standards such as those finalized by NIST introduces significant computational overhead, larger key and signature sizes, and complex integration challenges within blockchain consensus mechanisms and smart contract environments. Furthermore, the decentralized and immutable nature of blockchain creates unique challenges for cryptographic agility and key migration that are absent in centralized identity systems. This thesis addresses the urgent need for a purpose-built, quantum-resistant cryptographic framework specifically designed for the architectural and operational constraints of blockchain-based identity ecosystems.
11	**SVIT PS 11**	The exponential growth in cyberattack sophistication, frequency, and stealth has outpaced the predictive capacity of classical machine learning models deployed in threat intelligence platforms. Classical ML models are constrained by the curse of dimensionality when processing high-volume network traffic data, exhibit slow retraining cycles against zero-day threats, and fail to model complex non-linear attack correlations at scale. Quantum machine learning (QML) offers the potential to process high-dimensional cybersecurity feature spaces more efficiently through quantum kernel methods and variational quantum circuits; however, the practical application of QML to real-world threat intelligence pipelines remains nascent and largely theoretical. Key challenges include mapping classical cybersecurity datasets onto quantum feature spaces, handling noisy quantum hardware in inference pipelines, and achieving prediction latency compatible with real-time threat response requirements.
12	**SVIT PS 12**	Decentralized Finance (DeFi) platforms operate through deterministic smart contracts that lack the capacity for contextual reasoning, adaptive responses to market dynamics, or autonomous risk management. The rigidity of current smart contract logic makes DeFi protocols vulnerable to flash loan attacks, oracle manipulation, and cascading liquidations events that classical rule-based contracts cannot preemptively counter. While artificial intelligence offers dynamic decision-making capabilities, its integration with blockchain smart contracts faces fundamental challenges, including the on-chain computational cost of AI inference, the non-determinism of ML models conflicting with blockchain consensus requirements, and the oracle problem for feeding real-world data to smart contracts. The absence of intelligent, self-governing DeFi protocols creates systemic financial risk and limits the scalability and user adoption of decentralized financial infrastructure.
13	**SVIT PS 13**	Web3 applications built on public blockchain infrastructure require all transaction data to be broadcast and verified by every network participant, creating an irreconcilable tension between the blockchain's transparency mandate and users' legitimate privacy expectations. Existing privacy solutions such as coin mixers and confidential transaction schemes offer limited privacy guarantees, are computationally expensive, and fail to generalize across diverse Web3 application domains including NFTs, decentralized social networks, and on-chain governance. Zero-knowledge proofs (ZKPs) offer a mathematically rigorous framework for proving computational claims without revealing underlying data; however, their practical deployment in Web3 contexts is hindered by high proof generation costs, circuit complexity, limited developer tooling, and poor scalability under high transaction volumes. This thesis investigates the design of

		efficient, general-purpose ZKP frameworks optimized for Web3 scalability and cross-application privacy.
14	**SVIT PS 14**	Data sharing across organizational boundaries is essential for training high-quality machine learning models in domains such as healthcare, finance, and supply chain management; however, data sovereignty concerns, competitive pressures, and regulatory constraints such as GDPR severely restrict direct data exchange. Federated learning mitigates raw data exposure but remains vulnerable to gradient inversion attacks, model poisoning by malicious participants, and a lack of transparent contribution accounting. Centralized federated learning aggregation servers represent critical trust bottlenecks and points of failure. Blockchain technology offers decentralized governance, immutable audit trails, and cryptographic accountability; however, integrating blockchain with federated learning introduces latency challenges, smart contract computational limits, and incentive design complexities. This research addresses the need for a trustworthy, incentive-compatible, and attack-resilient decentralized data sharing ecosystem.
15	**SVIT PS 15**	The contemporary cyber threat landscape is characterized by advanced persistent threats (APTs), polymorphic malware, and living-off-the-land attack techniques that evade signature-based and rule-driven detection systems. Static security policies and perimeter-based defense architectures are fundamentally inadequate against adversaries who continuously adapt their tactics, techniques, and procedures (TTPs) to bypass known defenses. Behavioral threat modeling offers a more resilient detection paradigm; however, its practical implementation requires processing high-dimensional, high-velocity telemetry data in real time, a task that overwhelms conventional SIEM platforms. AI-driven approaches can model complex behavioral patterns but suffer from high false positive rates, poor generalization to novel attack classes, and a lack of adaptive response mechanisms that can modify defense configurations in reaction to detected threats. This thesis addresses the need for a dynamic, self-adapting cyber defense system grounded in behavioral AI and threat intelligence.
16	**SVIT PS 16**	Traditional perimeter-based security models operate on the flawed assumption that all entities within a network boundary are trustworthy, a model that is wholly inadequate in distributed cloud environments, remote work paradigms, and supply chain ecosystems. Zero-trust security architecture (ZTA) mandates continuous verification of every user, device, and session; however, current implementations rely on binary authentication events and static access policies that fail to adapt to contextual risk factors such as anomalous behavioral shifts, compromised credentials, or insider threat scenarios. Password-based and token-based authentication mechanisms are susceptible to credential theft, replay attacks, and social engineering. Biometric AI models offer continuous, non-intrusive behavioral authentication but face challenges of presentation attack detection, drift in biometric templates, and privacy implications of continuous physiological monitoring.
17	**SVIT PS 17**	Deep learning models deployed in safety-critical applications such as autonomous driving, medical diagnosis, and facial recognition are alarmingly vulnerable to adversarial examples of carefully crafted input perturbations that cause models to produce incorrect outputs with high confidence. The adversarial robustness problem represents a fundamental scientific challenge, as adversarial perturbations exploit geometric properties of high-dimensional loss landscapes in ways that are invisible to human perception but catastrophic in consequence. While numerous defense strategies have been proposed including adversarial training, certified defenses, and input preprocessing no existing approach

		provides uniform robustness across attack types, architectures, and domains without incurring significant performance penalties on clean data. The absence of a comprehensive, adaptive defense framework that generalizes across white-box, black-box, and transfer attacks severely limits the deployment of deep learning in adversarial environments.
18	**SVIT PS 18**	Chronic disease management and post-operative patient monitoring increasingly demand continuous, real-time physiological data analysis; however, cloud-centric health monitoring architectures introduce unacceptable communication latency, network dependency, and data privacy risks that are incompatible with life-critical clinical applications. Wearable IoT devices offer ubiquitous sensing but are constrained by severe computational limitations, battery capacity, memory, and intermittent connectivity. Existing on-device AI inference approaches fail to achieve the accuracy-efficiency trade-off required for high-stakes healthcare decisions, and most models are not designed for the distributional heterogeneity inherent in physiological data across diverse patient populations. Additionally, the lack of real-time anomaly detection for cardiac arrhythmias, hypoglycemia, and neurological events on edge devices poses a significant clinical safety risk.
19	**SVIT PS 19**	Smart city infrastructure generates petabytes of heterogeneous data daily from sensors embedded in traffic systems, environmental monitors, utility grids, and public safety networks. Centralized cloud processing of this data introduces prohibitive communication costs, unacceptable response latency for time-sensitive urban services, and massive energy consumption from data transmission and large-scale data center operations. Existing edge computing deployments lack intelligent task scheduling, workload migration, and resource orchestration capabilities necessary to dynamically balance computational load across geographically distributed edge nodes. Furthermore, individual edge devices operate in isolation without collaborative learning mechanisms, limiting their adaptive intelligence. The absence of energy-aware distributed AI frameworks that coordinate urban edge nodes results in resource underutilization, service degradation, and unsustainable energy expenditure incompatible with smart city sustainability goals.
20	**SVIT PS 20**	Large-scale IoT deployments in industrial, agricultural, and critical infrastructure contexts comprise thousands of heterogeneous devices that are inherently prone to hardware failures, software faults, cyberattacks, and environmental disruptions. Manual fault detection and recovery processes are infeasible at this scale and introduce dangerous service gaps in safety-critical systems. While AI-based fault detection has shown promise, autonomous recovery without human intervention remains largely unsolved, particularly in adversarial environments where malicious actors may deliberately induce system faults to exploit recovery mechanisms. The lack of tamper-resistant audit trails for fault events and recovery actions creates accountability gaps in critical infrastructure operations. Integrating self-healing autonomy with blockchain-based trust mechanisms offers a path toward resilient, accountable, and autonomous IoT ecosystems, but the architectural, computational, and consensus challenges of this integration have not been adequately addressed.
21	**SVIT PS 21**	Traditional educational delivery systems both physical and e-learning platforms employ a one-size-fits-all instructional approach that fails to accommodate the diverse cognitive styles, learning paces, and knowledge states of individual learners. Static educational content does not respond to learner confusion, disengagement, or mastery in real time, resulting in suboptimal learning outcomes and high attrition rates. Augmented and virtual reality (AR/VR) technologies

		offer immersive, experiential learning environments that significantly enhance knowledge retention and learner engagement; however, current AR/VR educational applications lack the artificial intelligence necessary to personalize content sequencing, difficulty adaptation, and instructional scaffolding based on real-time learner state estimation. The integration of affective computing, gaze tracking, and adaptive instructional design within XR environments remains a significant research frontier.
22	**SVIT PS 22**	Industrial operations in manufacturing, energy, and infrastructure sectors are characterized by complex, dynamic systems whose safe and efficient operation demands sophisticated monitoring, predictive maintenance, and real-time intervention capabilities. Conventional digital twin implementations provide virtual replicas of physical assets but lack immersive visualization interfaces and AI-driven analytical intelligence, limiting their utility for human operators who must rapidly interpret multi-dimensional system states under time pressure. Extended reality (XR) technologies offer the potential for intuitive, spatially anchored interaction with digital twin data; however, current XR-digital twin integrations suffer from high latency in state synchronization, the absence of AI-based predictive analytics, limited multi-user collaborative support, and insufficient fidelity for safety-critical simulation scenarios.
23	**SVIT PS 23**	Social networks and biological systems including protein-protein interaction networks, gene regulatory networks, and epidemic propagation graphs exhibit rich, non-Euclidean structural properties that cannot be adequately captured by traditional machine learning models designed for grid-structured or tabular data. Conventional deep learning architectures fail to model the relational dependencies, community structure, temporal dynamics, and heterogeneity of nodes and edges inherent in real-world complex networks. While Graph Neural Networks (GNNs) offer a native representation for graph-structured data, current GNN architectures face critical limitations including over-smoothing in deep networks, scalability challenges on massive graphs, limited expressiveness beyond the Weisfeiler-Leman isomorphism test, and poor generalization to evolving dynamic graphs. The absence of GNN frameworks purpose-built for the unique structural and temporal properties of social and biological networks limits their predictive and explanatory capacity.
24	**SVIT PS 24**	Modern medical imaging modalities including high-resolution MRI, CT, PET, and whole-slide digital pathology generate exceptionally high-dimensional image data that imposes critical storage, transmission, and computational burdens on healthcare information systems. Traditional compression standards such as JPEG2000 and HEVC are not designed to preserve diagnostically critical features in medical images and can introduce clinically significant artifacts at high compression ratios. AI-based compression approaches using variational autoencoders and neural codecs offer superior reconstruction quality but have not been adequately optimized for the specific semantic requirements of medical image analysis, the need for region-of-interest preserving compression, or compliance with clinical and regulatory standards for diagnostic image fidelity.
25	**SVIT PS 25**	Contemporary microservices architectures decompose software systems into independently deployable services; however, their orchestration remains predominantly static, relying on manually configured service meshes, rule-based load balancers, and reactive auto-scaling policies that are unable to reason about service interdependencies, user intent, or emergent system behaviors. As microservices ecosystems grow in scale and complexity, the operational burden of configuration management, fault diagnosis, and performance optimization grows

		exponentially. The absence of cognitive intelligence in microservice orchestration results in resource inefficiency, poor resilience to cascading failures, and inability to adapt architectural configurations to shifting workloads and business context. This thesis envisions and constructs cognitive microservices a new paradigm in which AI-driven intelligence is embedded within the orchestration layer to enable self-configuration, self-optimization, self-healing, and self-protection of distributed software architectures.
26	**SVIT PS 26**	Design and prototype an AI agent system that continuously monitors network logs from a simulated hybrid cloud environment, detects anomalous patterns that could indicate a cyberattack (including AI-generated synthetic attacks), classifies threat severity, and automatically triggers a structured incident response workflow all without human intervention for the first 60 seconds of detection.
27	**SVIT PS 27**	Build an AI agent system that conducts a 15-minute diagnostic assessment for an engineering student, identifies specific knowledge gaps across their core subjects, generates a personalised 30-day learning path using only freely available resources (YouTube, NPTEL, GitHub, documentation), tracks daily progress, and adapts the plan based on what the student completes and what they struggle with.
28	**SVIT PS 28**	Build an AI-powered medical triage agent that can collect patient symptoms through a conversational interface, ask clarifying questions using a structured clinical reasoning pattern, classify the urgency level (self-care / visit clinic within 48 hours / go to hospital now / call emergency), provide basic guidance for self-care cases, and refer to the nearest relevant healthcare facility. The system must operate safely with explicit guardrails preventing it from diagnosing or prescribing.
29	**SVIT PS 29**	Design an AI agent system that monitors publicly available signals (news, shipping data, weather, geopolitical events) for a specific supply chain segment (choose one: electronics, pharmaceuticals, automotive parts, or textiles), detects early warning signals of potential disruption, quantifies the potential impact on an MSME's inventory and delivery commitments, and recommends specific mitigation actions with time estimates.
30	**SVIT PS 30**	Build an AI agent that accepts a contract document (PDF or text paste), identifies high-risk clauses using a structured risk framework, explains each risk in plain language that a non-lawyer can understand, scores the overall contract risk level, and suggests specific renegotiation language for the top 3 risks. The system must include clear disclaimers and must not provide legal advice.
31	**SVIT PS 31**	Build an AI advisory agent that a farmer can interact with (text or voice description) to describe crop symptoms, receives a structured analysis of likely diseases or deficiencies, gets specific actionable advice in simple language (including which locally available treatments to try), receives a weather-integrated recommendation (e.g., 'do not spray pesticide today rain expected in 6 hours'), and gets an escalation path when the issue requires expert intervention.
32	**SVIT PS 32**	Build an AI agent system that analyses an employee's current skill profile (from resume or self-assessment), maps it against a set of open internal roles and required skill profiles, identifies the exact skill gaps for each potential role, recommends a specific upskilling path for the top 2 matched roles using free resources, and generates a 'career mobility report' the employee can take to their manager.

33	**SVIT PS 33**	Build an AI agent that a small merchant can use to: analyse incoming UPI transaction patterns in real time, flag transactions that match known fraud patterns (urgency pressure, unusual amounts, unknown senders, late-night transactions), explain WHY a transaction is flagged in simple language, recommend immediate action, and log flagged transactions for dispute filing.
34	**SVIT PS 34**	Build an AI agent system that helps an IT company track carbon emissions at the project level (data centre usage, employee commute, business travel, cloud compute), calculates total Scope 1, 2, and 3 emissions using standard GHG Protocol methodology, benchmarks against industry standards, identifies the top 3 reduction opportunities with quantified impact, and generates a report in the format required for GRI (Global Reporting Initiative) disclosure.
35	**SVIT PS 35**	Build an AI agent system that receives citizen complaints (text description), automatically categorises the complaint into the correct department (roads, sanitation, water, electricity, parks, illegal construction), checks for duplicate complaints about the same location/issue, assigns a priority score based on severity and number of affected residents, drafts an acknowledgement message to the citizen, and generates a structured work order for the relevant department.
36	**SVIT PS 36**	Build an AI Ethics Audit agent that analyses an AI system's documentation (technical specification, intended use case, training data description, deployment context) and produces a structured ethics audit report covering: bias risk assessment, transparency score, accountability chain, potential for misuse, alignment with EU AI Act risk categories, and recommended safeguards. The system must reason transparently and cite specific evidence from the provided documentation for every finding.
37	**SVIT PS 37**	Build an AI Code Review agent that accepts a code snippet or file, detects both standard vulnerabilities AND vulnerabilities specifically common in AI-generated code (hallucinated API calls, deprecated function usage, logic errors that pass syntax checks, insecure default configurations), classifies each finding by severity (Critical / High / Medium / Low), explains the vulnerability in plain language, provides a corrected code snippet for each Critical and High finding, and estimates the OWASP Top 10 category for security findings.
38	**SVIT PS 38**	This problem statement proposes the development of a Smart Health Surveillance and Early Warning System that can detect, monitor, and help prevent outbreaks of water-borne diseases in vulnerable communities. The system can be: • Collect health data from local clinics, ASHA workers, and community volunteers via mobile apps or SMS. • Use AI/ML models to detect patterns and predict potential outbreaks based on symptoms, water quality reports, and seasonal trends. • Integrate with water testing kits or IoT sensors to monitor water source contamination (e.g., turbidity, pH, bacterial presence). • Provide real-time alerts to district health officials and local governance bodies. • Include a multilingual mobile interface for community reporting and awareness campaigns. • Offer dashboards for health departments to visualize hotspots, track interventions, and allocate resources.
39	**SVIT PS 39**	Image based breed recognition for cattle and buffaloes of India • Uses Artificial Intelligence (AI) and image analysis to recognize and classify the breed of cattle and buffaloes based on images. • Can handle diverse environmental backgrounds, lighting conditions, and varying animal poses.

		• Maintains a breed database (for the most common Indian cattle and buffalo breeds and their crosses). • Provides breed suggestions or confirmations at the time of registration in BPA. • Can be seamlessly integrated with the BPA platform to support real-time validation or correction of breed entries. • Includes a user-friendly interface for FLWs with minimal technical training requirements.
40	**SVIT PS 40**	Al-Based Crop Recommendation for Farmers A mobile-based prototype offering farmers customized, science-guided crop advice, increasing income, making resources more efficient, and facilitating sustainable agriculture. The solution should deliver an AI-powered platform that integrates text and image-based interactions tailored for agricultural use. It must support voice and chat interfaces in local languages, enabling farmers to ask questions and receive actionable responses.
41	**SVIT PS 41**	AI-Driven Crop Disease Prediction and Management System. Crop diseases can devastate yields, leading to significant financial losses for farmers. Early detection and timely intervention are crucial for effective management. Description: Develop an AI-driven system that analyzes crop images and environmental data to predict potential disease outbreaks. This system will provide farmers with actionable insights and treatment recommendations to mitigate risks. Expected Solution: A mobile and web-based application that utilizes machine learning algorithms to identify crop diseases and suggest preventive measures and treatments based on real-time data.
42	**SVIT PS 42**	Gamification of Anti-Doping Information.Traditional methods of disseminating anti-doping information often fail to engage younger athletes effectively. Gamification, the application of game-design elements and game principles in non-game contexts, can be a powerful tool to make learning about anti-doping rules more engaging and interactive. By incorporating gamification, the National Anti-Doping Agency can enhance its educational initiatives and foster a culture of clean sport among young athletes. Description: The above problem statement envisages the creation of a gamified platform to educate athletes on anti-doping rules and practices. The solution should be engaging, informative, and foster a culture of clean sport among young athletes. It should incorporate game elements such as points, badges, leader boards, and challenges to motivate users to learn and apply anti-doping information. Key Objectives (a) Engagement: Develop engaging and interactive content that motivates athletes to learn about anti-doping, leveraging the appeal of gaming to enhance educational outcomes. (b) Education: Ensure that the gamified content effectively conveys critical anti-doping information, covering rules, prohibited substances, testing procedures, and the importance of clean sport. (c) Behavioural Change: Encourage positive attitudes and behaviours towards clean sport, promoting a sense of responsibility and commitment among young athletes. Detailed Requirements (a) Platform Development Develop a gamified mobile app and web-based platform that is accessible to athletes across different devices. Ensure the platform is user-friendly, visually appealing, and easy to navigate. (b) Game Elements Incorporate points, badges, leader boards, and levels to incentivize participation and learning. Design challenges, quests, and mini-games that teach anti-doping concepts in an engaging way. Include social features that allow users to compete with friends, share achievements, and collaborate on challenges. (c) Educational Content Develop comprehensive educational modules covering

		various aspects of anti-doping. Use multimedia content, including videos, animations, and interactive infographics, to make learning enjoyable. Include real-life scenarios and case studies to provide practical insights and reinforce learning. (d) Assessment and Feedback Implement quizzes and assessments to test users' knowledge and understanding of anti-doping rules. Provide immediate feedback and explanations to help users learn from their mistakes. Offer rewards and recognition for users who demonstrate high levels of knowledge and commitment. (e) Analytics and Reporting Integrate analytics tools to track user engagement, progress, and learning outcomes. Generate reports on user performance, highlighting areas of strength and areas needing improvement. Use data insights to continuously improve the platform and its content. Expected Solution: A highly engaging and educational gamified platform for anti-doping information. Increased knowledge and awareness of anti-doping rules and practices among young athletes. Positive behavioural changes towards clean sport, with athletes demonstrating greater commitment to anti-doping principles. Enhanced engagement and participation rates in anti-doping education programs.
43	**SVIT PS 43**	Development of an educational game (web-based and mobile- based) on groundwater conservation and management. Learning while you play is considered the most effective way of teaching. Internet/mobile based games could be one of the best ways to lure school kids, youth and water enthusiasts to learn the nuances of ground water management. With this backdrop it is proposed to develop an internet/mobile based game that teaches good practices in groundwater conservation in an interactive and fun way. The game should take into account various interventions (artificial recharge, microirrigation, crop diversification) and possible scenarios (drought, surplus rain, contamination etc). The gamer can earn points or coins based on the choices that he (or she) makes. The scores of registered gamers will be stored online and water-smart youths can be identified and certified. The game can be used for training and awareness creation.
44	**SVIT PS 44**	Forecasting Future Water Requirements and Assessing Storage Capacities in Reservoirs. Water resources are fundamental to sustaining human life, agriculture, industry and ecosystem. Accurate forecasting of future water requirements along with the assessment of current storage capacities are crucial for effective water resources management and planning. With a growing population. The domestic, Industrial, Agricultural and ecological demands increase, which in turn leads to strain on existing water Infrastructure. Description Analyzing the storage capabilities of the reservoirs in the country is essential. It is also essential for the policy makers to plan for a sustainable water resource management. The challenge involves creating a predictive model that can accurately forecast the future water requirements and evaluate the current and future storage capabilities. The model must integrate various data sources, including historic water usage, climate projections, population growth trends, and agricultural practices. It should also account for variability in climate conditions, demographic shifts and changes in land use. A strategic planning and decision-making support system can provide clear insights into future water demand and storage needs, by identifying potential risks. Expected Solution It is expected to assess the existing and future storage capabilities, identify the anticipated conditions and impact on the storage capabilities. Identify the potential gaps where additional storage and infrastructure capabilities are needed. It is also required to create a scenario taking into account the extreme weather events and population demands. The Policy makers should have an insight of water management issues for providing sustainable water management. A strategy should be evolved for enhancing water storage capacities, optimizing water usage and mitigating identifying risks.

45	**SVIT PS 45**	Real-Time Fraud Detection with Grounded AI Explanations Problem Statement: Design a real-time fraud detection system that identifies suspicious financial transactions using machine learning or deep learning techniques. The system should generate human readable explanations using generative AI, ensuring that outputs are grounded in model predictions and not hallucinated. It must handle real-world challenges such as imbalanced datasets and evolving fraud patterns. The solution should be efficient for real-time use and provide reliable, interpretable results for decision-making. Constraint: Generative AI outputs must be verifiable and aligned with model predictions.
46	**SVIT PS 46**	Multi-Modal Fake News & Deepfake Detection with Evidence Generation Problem Statement: Develop a multi-modal AI system that detects fake news by analyzing text, images, and metadata using deep learning techniques. The system should leverage generative AI to produce evidence-based explanations or corrected summaries, ensuring outputs are grounded in detected features. It must handle adversarial or manipulated content and provide reliable credibility assessment. The solution should be robust, scalable, and suitable for real-world misinformation detection. Constraint: Generated explanations must reference input data or model outputs, not generic reasoning
47	**SVIT PS 47**	Personalized Learning System with Grounded AI Tutor Problem Statement: Create an intelligent learning system that predicts student performance and dynamically generates personalized learning paths using machine learning techniques. The system should integrate a generative AI tutor to provide explanations, quizzes, and guidance, ensuring responses are grounded in curriculum or learning data. It must adapt continuously based on student progress and provide meaningful recommendations. The solution should enhance learning outcomes through reliable and interpretable AI support. Constraint: AI tutor responses must be grounded in data and avoid hallucinated content.
48	**SVIT PS 48**	AI Code Reviewer with Auto-Fix Validation Problem Statement: Develop an AI-powered code review system that identifies bugs, inefficiencies, and code quality issues using machine learning or NLP techniques. The system should use generative AI to suggest fixes and improvements, ensuring that generated solutions are context-aware and executable. It must validate suggested fixes through testing or analysis and provide meaningful feedback on code quality. The solution should support developers in improving reliability and efficiency. Constraint: Generated fixes must be validated and functionally correct
49	**SVIT PS 49**	Healthcare Risk Prediction with Reliable AI Assistant Problem Statement: Build a healthcare AI system that predicts patient risk levels using incomplete and noisy medical data through machine learning or deep learning techniques. The system should generate clear, doctor-friendly insights using generative AI, ensuring outputs are grounded in patient data and not hallucinated. It must handle uncertainty, ensure fairness, and provide reliable decision support. The solution

		should be robust and applicable in real-world healthcare scenarios. Constraint: AI-generated insights must be accurate, explainable, and data-driven.
50	**SVIT PS 50**	Intelligent Document Verification with AI Reasoning Problem Statement: Develop an AI system that verifies the authenticity of documents (e.g., IDs, certificates, invoices) using computer vision and machine learning techniques. The system should detect tampering, inconsistencies, or forged elements and use generative AI to provide clear reasoning behind its decisions. It must handle varied formats, low-quality inputs, and partial data. The solution should be robust, scalable, and applicable to real-world verification workflows such as banking or recruitment. Constraint: Generated explanations must be grounded in detected document features and not generic.
51	**SVIT PS 51**	Predictive Maintenance System with AI Failure Explanation Problem Statement: Design a predictive maintenance system that forecasts equipment or machine failures using time-series and sensor data. The system should leverage machine learning or deep learning models to detect anomalies and predict failures in advance. It must use generative AI to explain the cause of predicted failures in a human-understandable way. The solution should handle noisy data, adapt to changing conditions, and support real-time monitoring. Constraint: AI-generated explanations must be traceable to sensor patterns or model outputs.
52	**SVIT PS 52**	Problem Statement: Build a system that analyzes customer behavior using transaction, interaction, and activity data to predict churn, preferences, or engagement patterns. The system should use machine learning models for prediction and generative AI to produce actionable business insights and recommendations. It must handle large-scale, noisy data and adapt to evolving user behavior. The solution should provide interpretable and practical outputs for decision-making. Constraint: Generated insights must be data-driven and aligned with model predictions, avoiding generic suggestions.
53	**SVIT PS 53**	Software for Speech Language Therapy Clinical Services. The rehabilitation institutes involved in this activity face high therapy cases load and more number of student therapists with less number of therapy supervisors. Following this system in manual mode impedes providing prompt services. Description: Clinical services with respect to speech language therapy begins with patient allocation to the therapist. The therapist then prepares a therapy plan with goals and activities. The respective case supervisor needs to check the therapy plan and provide the inputs for further continuation of regular therapy sessions. At least post ten therapy sessions the therapist writes the progress report of the therapy case. Further, it is evaluated by the supervisor. This Cycle of speech language therapy clinical services continues until the therapy case has completely improved or discontinued due to various reasons. The supervisor based on overall handling of therapy sessions gives clinical ratings. Expected Solution: The process needs to become digitized with the use of software where in right from the allocation of cases, therapy related documentation by both therapist and supervisor, clinical evaluation of therapy,

		feedback and clinical rating can be performed.
54	**SVIT PS 54**	Use of Digital Technology to calculate Water Footprints for different Agricultural Products. The water footprint measures the amount of water used to produce each of the goods and services we use. The water footprint helps us understand for what purposes our limited freshwater resources are being consumed. The impact of it depends on where the water is taken from and when, if it comes from a place where water is already scarce, the consequences can be significant and require action. Detailed Description The increase in the amount of non-available water due to pollution and scarce groundwater level has added more water footprints, at the community as well as at the personal levels. An increased water footprint directly affects the health and future of the citizens. Preventing severe drought in water-stressed areas is only going to be possible if water is used with more care and efficiency, this can be achieved if we have readily available data of water footprints. Expected Solution Hence, by using digital technologies like AI, Big Data, Block chain etc. and computer languages, a user friendly app or website may be developed which can provide the water footprints of different items/ final products we eat by feeding little inputs like name, or just by scanning through camera like Google lens. The app should support local languages; this will ensure pan India usage and sensitize the people about water footprints of items they use in daily life.
55	**SVIT PS 55**	Fake social media accounts and their detection. At present the ITBP guards 3,488 km long India-China borders ranging from the Karakoram Pass in Ladakh to Jachep La in Arunachal Pradesh. Apart from this, the Force also has important roles in many internal security duties and operations against the left wing extremism in the state of Chhattisgarh. Creating fake accounts on Facebook, Instagram or at any other platform and fake account uses, should be identified for account suspension or legal action. To safeguard the organization from the unknown fake account messages over any social sites, a tool may be developed for their identification. Also a central agency should be designated to get the information about the identified fake account holder informed by the developed tools and accordingly, concerned social site organizations will approach to delete/suspend the fake account in a time bound manner worldwide. Description: How to discover/identify fake profiles on Facebook, Instagram, twitter or other social apps using tools. Also subsequently how to ignore/report/make to delete of these identified fake accounts by the tools/ through a concerned agency in India. Expected solution: 1. Tools for identifications of fake accounts should be developed for popular social sites. 2. A Central Agency should be designated to get the information about the identified fake account holder info by the developed tools and accordingly, concerned social site organizations will approach to delete/suspend the fake accounts in time bound manner worldwide.
56	**SVIT PS 56**	Recovery of Deleted Data and Associated Metadata from XFS and Btrfs Filesystems. Digital evidence has become increasingly crucial in forensic investigations. The recovery of deleted data from storage devices is essential for reconstructing timelines, identifying suspects, and uncovering critical information. Traditional file systems like FAT and NTFS have been extensively studied, and tools for recovering deleted data from them are relatively mature. However, modern file systems like XFS and Btrfs, designed for performance and reliability, employ complex data structures that pose significant challenges for data recovery. Forensic investigations often involve recovering various file types, including documents files, log files, and system files. These files contain valuable information about system activities, user behaviour, and potential criminal activities. The ability to recover deleted files along with their complete metadata,

		such as creation, access, modification, and deletion timestamps, is crucial for establishing timelines and corroborating evidence. Detailed Description :XFS and Btrfs file systems offer advanced features like journaling, copy-on-write, and efficient data allocation. While these features enhance system performance and data integrity, they also complicate the process of recovering deleted data. When a file is deleted in these file systems, the data itself is not immediately erased; instead, the file system marks the allocated blocks as free for reuse. This delayed overwriting of data presents an opportunity for recovery, but it also requires specialized techniques to extract and analyse the deleted data. Moreover, recovering accurate metadata associated with deleted files is equally challenging. Metadata is critical for establishing the context of the recovered data and determining its relevance to the investigation. Extracting metadata from XFS and Btrfs file systems requires a deep understanding of their internal structures and data allocation mechanisms. Expected Solution: An ideal solution would be a comprehensive data recovery technique specifically designed for XFS and Btrfs file systems. These techniques should be able to: 1. Efficiently recover deleted data: Develop algorithms and techniques to identify and extract deleted files from the complex data structures of XFS and Btrfs. 2. Support a wide range of file types: Recover Text-Based Document Formats(doc,docx, rtf, pdf, txt, odt, html, xml, ppt, odp, xls, ods, log, csv, tsv, txt, conf, ini, cfg etc), archives file(zip, tar, rar, iso, rpm, deb etc), Image-Based Document Formats(jpg, jpeg, png, gif, tif etc), executables binaries(.elf, .so, .a, exe, dll, bat, cmd) scripts files(ps, ps1, sh, bash, zsh, py etc), database file(.db etc) and other relevant data formats. 3. Extract complete metadata: Recover accurate creation, access, modification, and deletion timestamps, file names, and other essential metadata associated with deleted files. 4. Provide user-friendly interface: Offer an intuitive interface (GUI/CLI) for easily navigating recovered data and generating reports. 5. Ensure data integrity: Implement robust data validation and verification mechanisms to maintain the integrity of recovered data.
57	**SVIT PS 57**	Creating a cyber triage tool to streamline digital forensic investigation. To design and develop an innovative digital forensics and incident response tool with an intuitive and accessible interface for investigators, that streamlines the process of importing evidence, conducting automated analysis, and generating detailed reports. The tool should feature an interface with clear navigation & real-time data visualization and should support: 1. Automated data collection from RAW images (forensic images) and other formats using disk imaging tools 2. Automate the scanning and analysis of data, including files, system logs, registry entries, network activity etc. 3. Identify indicators of compromise (IOCs) and related suspicious activities 4. Integrate AI/ML algorithms for anomaly detection and pattern recognition. The AI/ML feature should incorporate a scoring system and recommendation engine that allow investigators to quickly focus on the important artifacts. 5. User-friendly review options should include interactive timelines and graphical summaries, while comprehensive reporting capabilities should allow exports in various formats such as PDF, JSON, and CSV.
58	**SVIT PS 58**	Centralized application-context aware firewall. Develop an application firewall for end-points that can identify and restrict access of applications to external networks/hosts. The application firewall should provide further granular control of restricting domains, IP addresses and protocols for each application. The firewall should be manageable through a centralized web console where policies for each end-point and application can be centrally deployed. Firewall agents should also be able to monitor network usage behaviour of each application and generate alerts on the central dashboard for any traffic anomaly using AI/ML. Challenge: Applying separate firewall policies for each application running on the

		end-point and managing them through a central web console. Usage: End-point security, network security Users: Cyber security teams Available Solutions (if Yes, reasons for not using them): Individual components are available Desired Outcome: The solution should provide following components: 1. Solution should identify the domains and protocols that any application is trying to access. Further, it should enable any such network traffic which is not already allowed via centralized console. 2. Context-aware application firewall agent that shall manage firewall policies for each application running on end-point. The agent shall also collect network usage logs of each application and send it to the central server. 3. Central web management console that shall be able to manage all end-points and applications 4. The solution should work for Windows end-points. Bonus points for Linux 5. Solution should also detect abnormal network behaviour of applications.
59	**SVIT PS 59**	Software solution to identify the end receiver of a cryptocurrency transaction. Use of cryptocurrencies like bitcoin, USDT, Monero etc. for drug trafficking activities are increasingly becoming common. The relative anonymity and speed provided by cryptocurrencies are misused by drug traffickers as a mode of transaction for drug sales and also as an asset to amass the proceeds of crime. Description: Drug traffickers operating on darknet and elsewhere on internet enabled platforms like social media apps, messaging services etc receive the value of drugs through cryptocurrencies. In a few cases, the drug traffickers save their proceeds of crime in cryptocurrencies. During the course of investigations and also through intelligence, wallet addresses and transactions hash related to drug transactions are obtained by Drug Law Enforcement Agencies, Further, it has also been observed that these finds are often passed through a variety of services like tumblers, mixers, bridges etc to further anonymize the transactions. It is important to follow the trial of funds associated with drug transactions to identify the real persons behind the drug trafficking network. Expected Solution: Development of a software solution to follow the cryptocurrency transaction trial associated with a wallet id/transaction to find out the real receiver of the funds in a drug related transaction.
60	**SVIT PS 60**	Software solutions to identify users behind Telegram, WhatsApp and Instagram based drug trafficking. Use of encrypted messaging/social media apps like Telegram, WhatsApp and Instagram for drug trafficking are on the rise. Channels operating on Telegram and WhatsApp and Instagram handles are blatantly being misused by drug traffickers for offering various narcotic drugs and Psychotropic substances for sale. Description: WhatsApp and Telegram channels and Instagram handles are created by drug traffickers to offer various drugs for sale to their subscribers. Customized Telegram bots are also created by some of the drug traffickers to sell drugs. It is most worrisome that the majority of the drugs which are being offered on sale through Telegram, WhatsApp and Instagram are dangerous synthetic drugs like MDMA, LSD, Mephedrone etc. The above three apps are also used by drug traffickers for drug communication. Expected Solution: Development of a software solution to identify live Telegram and WhatsApp channels/bots and Instagram handles that are offering drugs for sale in India. Solution also should focus on triangulating identifiable parameters like IP address, mobile number, email id etc of the users behind the channel/bot/handle.
61	**SVIT PS 61**	DDoS Protection System for Cloud: Architecture and Tool. Many organizations are using Cloud for hosting their web applications. The attackers can try to attack these web servers for achieving Denial of Service attack. Distributed Denial-of-Service (DDoS) attack is a malicious attempt to disrupt the normal traffic of a targeted server, service or network of Cloud infrastructure by

		overwhelming the target or its surrounding infrastructure with a flood of internet traffic. DDoS attacks achieve effectiveness by utilizing multiple compromised computer systems as sources of attack traffic. Exploited machines can include computers and other networked resources. Therefore, it is essential to develop appropriate security tools to counter and protect against these attacks. • Description: The most obvious symptom of a DDoS attack is that a website or service suddenly becomes slow or unavailable. But since a number of causes such a legitimate spike in traffic can create similar performance issues, further investigation is usually required. Therefore, suitable analytics tools need to be developed to clearly identify an attack as DDoS. Following are some of the patterns for a DDoS attack: 1. Suspicious amounts of traffic originating from a single IP address or IP range 2. A flood of traffic from users who share a single behavioral profile, such as device type, geolocation, or web browser version 3. An unexplained surge in requests to a single page or endpoint 4. Odd traffic patterns such as spikes at odd hours of the day or patterns that appear to be unnatural (e.g. a spike every 10 minutes) There are other, more specific signs of DDoS attack that can vary depending on the type of attack. The tool developer should be creative to consider other signs also. For the above problem statements, following assumptions can be made: 1. Cloud is hosting a website and providing some services to its users. 2. The website should be always up and providing services to its users (high availability). 3. The attackers can flood the website directly or via other nodes (DDoS). 4. The attacker can also sabotage the link between a client and webserver. 5. The attack can come from outside or from within the cloud infrastructure. A solution needs to be built by suitably designing the cloud architecture and developing some tool (s) to automatically detect and recover from the DDoS attack. Expected Solution: A set of developed tool(s) along with a suitable Cloud architecture to be demonstrated. The demonstrated website should be protected well against different types of DDoS attack. In case of an attack, the developed security tools should be able to automatically detect and protect a website hosted on cloud infrastructure against DDoS attacks. The solution should also demonstrate the automatic recovery from the attack. As high availability is an essential feature, the down time (recovery time) should be minimized to the extent possible.
62	**SVIT PS 62**	Farmers Disease Diagnostic/Reporting Portal - Mobile Portal Al Based Agricultural productivity and food security are heavily dependent on the health of crops and livestock. Farmers, especially in remote or resource-limited areas, often face challenges in diagnosing and reporting diseases that affect their livestock and crops. Early diagnosis and timely reporting is crucial for effective disease management and prevention of widespread outbreaks. Description: A mobile portal powered by Artificial Intelligence (Al) for disease diagnosis and reporting can revolutionize how farmers manage animal and plant health. This portal enables farmers to use their smartphones to access diagnostic tools, report symptoms, and receive actionable advice. Al-algorithms can analyze reported data to provide accurate diagnoses and suggest appropriate treatments or interventions. This technology can empower farmers with the knowledge and tools to manage diseases more effectively, ensuring the health and productivity of their farms. Integrate Al-based software with existing NDLM to allow farmers to submit images and descriptions of disease signs and symptoms. The Al will generate suspected disease/condition reports with preventive measures and send alerts to veterinarians for appropriate action (Al-based software, Mobile application). Expected Outcomes . Enhanced Disease Diagnosis: Timely Reporting and Intervention: Increased Access to Expert Knowledge: Improved Farm Productivity. Data Collection and Analysis: Cost-Effective DiseaseManagement: Empowerment and Education of Farmers: Integration with

		Existing Surveillance Systems . Community Engagement and Support: Sustainable Agriculture Practices.
63	**SVIT PS 63**	Development of AI-ML based models for predicting prices of agri-horticultural commodities such as pulses and vegetables (onion, potato, onion). The Department of Consumer Affairs monitors the daily prices of 22 essential food commodities through 550 price reporting centres across the country. The Department also maintains a buffer stock of pulses, viz., gram, tur, urad, moon and masur, and onion for strategic market interventions to stabilize the volatility in prices. Decisions for market interventions such as release of stocks from the buffer are taken on the basis of the price trends and outlook. At present, the analyses of prices are based on the seasonality, historical and emerging trends, market intelligence inputs, crop sowing and production estimates. ARIMA based economic models have also been used to examine and forecast prices of pulses.
64	**SVIT PS 64**	Assured Contract Farming System for Stable Market Access. Farmers often face uncertainties in market access, leading to fluctuating incomes. Contract farming can provide stability by ensuring farmers have guaranteed buyers for their produce. Description: Develop a comprehensive platform that facilitates assured contract farming agreements between farmers and buyers. This platform will enable transparent communication, secure contracts, and timely payments, ensuring farmers have a reliable market for their crops. Expected Solution: An online marketplace that connects farmers with potential buyers, offering tools for contract management, price negotiation, and secure payment processing, thereby enhancing income stability and reducing market risks.
65	**SVIT PS 65**	Sustainable Fertilizer Usage Optimizer for Higher Yield. Excessive and improper use of fertilizers leads to soil degradation and reduced agricultural productivity, negatively impacting farmers' income. Description: Create a data-driven solution that recommends optimal fertilizer types and quantities based on soil health, crop type, and weather patterns, ensuring sustainable agricultural practices. Expected Solution: An application that analyzes soil data and provides tailored fertilizer recommendations, promoting sustainable farming while enhancing crop yield and farmer income.
66	**SVIT PSAR 66**	### **1. Machine Maintenance Prediction** **Problem Statement** Develop an intelligent **Predictive Maintenance System** for industrial machines that can accurately predict: - **Time until failure** (Remaining Useful Life / Time-to-Failure) - **Type or category of the expected failure** **Objective:** Build a Machine Learning / Deep Learning solution using time-series sensor data (vibration, temperature, pressure, current, RPM, etc.) and historical maintenance records to enable proactive maintenance, reduce unplanned downtime, and optimize maintenance costs. **Expected Outcomes:** - Accurate regression model for Time-to-Failure prediction - Classification model for failure type prediction - Dashboard/visualization showing predictions and confidence scores ---

67	**SVIT PSAR 67**	### **2. YOLO-based Highly Accurate Student Photo Detection from ID Cards** **Problem Statement** Build a robust computer vision application using **YOLO** (You Only Look Once) or any state-of-the-art object detection model to automatically detect, crop, and save student photographs from physical ID cards with very high accuracy. **Key Requirements:** - Reliably detect the photo region even if the ID card is **panned, tilted, rotated, or captured under poor lighting/angles**. - Automatically crop the photo cleanly (removing card borders and background). - Save the cropped photo in a folder with the **student's full name as the filename**. **Use Case:** Automation of student photo digitization for digital ID cards, attendance systems, or database management. ---
68	**SVIT PSAR 68**	### **3. Lightweight Person Detection System with Steady-State Signal** **Problem Statement** Design a **lightweight, real-time, and highly accurate person detection system** suitable for edge devices that outputs a stable binary signal indicating human presence. **Key Requirements:** - Output **True** when a person (man or woman) is detected and **maintain True** steadily as long as the person remains in the frame. - Switch to **False** only when no person is present and keep it **False** until a new person appears. - Minimize flickering or unstable state transitions. - Optimized for low computational resources (lightweight model). **Optional:** - Add classification (Man / Woman) **Use Cases:** Smart lighting, security systems, occupancy monitoring, energy management, and automation.